

MBA CORE ELECTIVE — FINANCIAL ANALYTICS

Reading the Market's Story — Trend, Noise, and the Limits of Prediction

3.5-HOUR WORKSHOP

Consulting-style, discussion-first — you will speak more than the professor.

NO STATS BACKGROUND NEEDED

Every idea starts with business intuition, then math, then Python.

FINAL CHALLENGE

Diagnose three mystery time series and defend your modeling choice.

Today's Workshop — A 3.5-Hour Learning Journey

Block 1 (45 min)

Why Time Series Matter & Types of Data

Block 2 (55 min)

Components, Decomposition & Predictability

Block 3 (70 min)

Random Walk Theory, Drift & Unit Roots

Block 4 (50 min)

Stationarity, Prices vs. Returns

Block 5 (30 min)

Business Implications & Techniques

Block 6 (20 min)

Final Consulting Challenge & Wrap-Up

Ground rule: this is a workshop, not a lecture. Expect a poll, debate, or activity every 8-10 slides — keep your phone/laptop handy for polls.

Can Anyone Beat the Stock Market?



Vote Now

- A. Yes — skilled analysts consistently beat the market.
- B. No — nobody can consistently beat a well-diversified index.
- C. Sometimes — luck and short-term skill exist, but not consistently.
- Vote with a show of hands (or the class poll link) before we discuss.



Why We Start Here

- Your answer to this question is really a hypothesis about how PREDICTABLE financial time series are.
- Everything in today's session — random walks, stationarity, drift — is really evidence for or against your vote.
- We will revisit this exact poll at the end of the session. Remember your answer.

Why Should a Manager Care About Time Series?

Three Decisions That Depend On It

- A CFO deciding when to hedge USD/INR exposure for an import contract.
- A portfolio manager deciding how much risk capital to allocate to equities vs. gold this quarter.
- An RBI policymaker deciding whether rising inflation is a trend or a temporary spike.

The Common Thread

- Every one of these decisions requires separating SIGNAL (a real, persistent pattern) from NOISE (random fluctuation).
- Get it wrong and you either overreact to noise (costly false alarms) or ignore a real signal (costly inaction).
- Time series analysis is the discipline of making that separation rigorously, not by gut feel.

What Patterns Do You See in NIFTY Over 5 Years?



 3 minutes: *THINK* alone (30 sec) — jot down 2 patterns. *PAIR* (90 sec) — compare with a neighbor. *SHARE* — 2-3 pairs report out.

NIFTY 50 — Five Years in One Chart

Illustrative price path built to match real index behavior: trend, crash, recovery, volatility



What Is a Financial Time Series?

Definition (Intuition First)

- A sequence of observations on the SAME variable, ordered in time, where the ORDER matters.
- Contrast: in a cross-sectional dataset, shuffling the rows changes nothing. In a time series, shuffling destroys the information.
- Financial examples: daily NIFTY close, monthly inflation, quarterly EPS, tick-by-tick trade prices.

Managerial Interpretation

- Because order matters, TODAY'S value is connected to YESTERDAY'S — this dependency is the entire reason time series methods differ from ordinary regression.
- Managers who treat time series data like cross-sectional data (e.g., averaging without regard to order) routinely misjudge risk and trend.

Four Types of Financial Time Series

Price Series

Levels of an asset over time: NIFTY close, HDFC share price, gold price. Usually trending and non-stationary.

Return Series

Percentage or log change period-to-period. The preferred series for modeling — usually stationary.

Volatility Series

A measure of dispersion over time (e.g., rolling standard deviation, VIX). Itself has time-series patterns — 'volatility clustering.'

Macro Series

Inflation, GDP growth, repo rate, UPI transaction volumes — usually monthly/quarterly, often seasonal.

Cross-Sectional vs. Time Series vs. Panel Data

Cross-Sectional

- Many entities, ONE point in time.
- Example: ROE of 50 NSE companies on 31-Mar-2024.
- Order of rows is meaningless.

Time Series

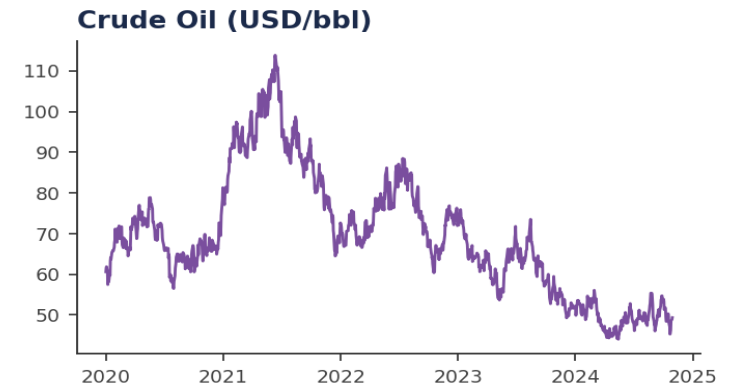
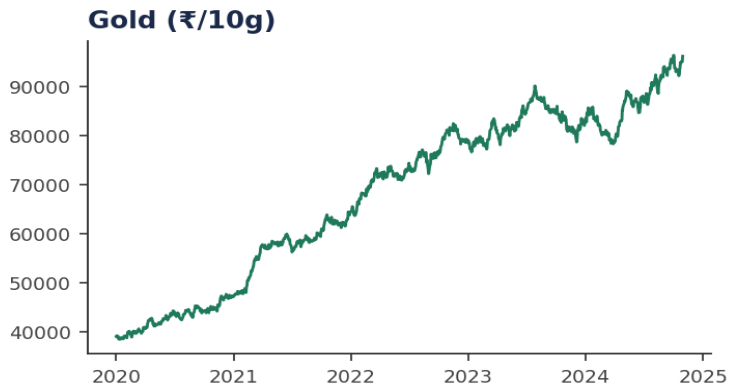
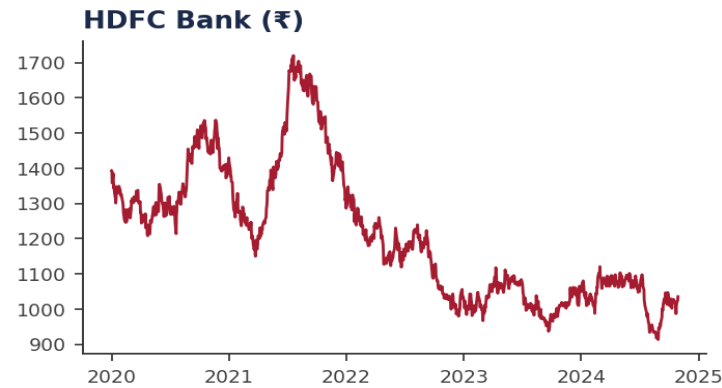
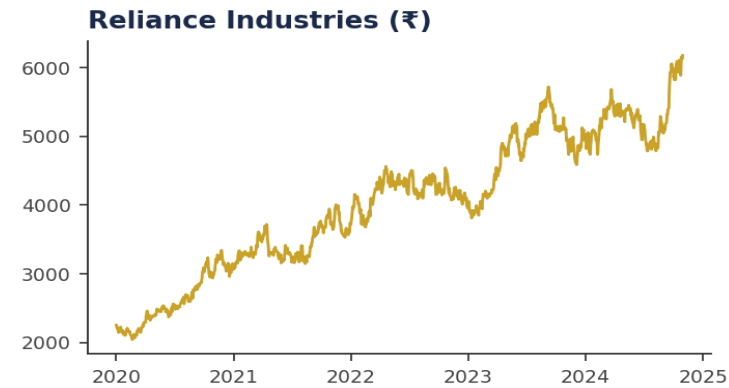
- ONE entity, observed repeatedly over time.
- Example: daily NIFTY close, 2020-2025.
- Order of rows is everything.

Panel Data

- MANY entities, observed repeatedly over time.
- Example: monthly stock returns for 50 companies over 5 years.
- Richest structure — combines both dimensions.

A Bloomberg-Style View — Six Markets, One Screen

Illustrative Market Dashboard — Six Financial Time Series



You Are an Equity Analyst at HDFC Securities



The Brief

- Your Head of Research asks: 'Pull together a one-page view of Reliance, HDFC Bank, and Gold for tomorrow's client call.'
- You have the dashboard from the last slide and nothing else.
- You have 5 minutes to prepare 3 talking points.



Discuss in Pairs

- What would you say about each series' trend?
- Which series would you be MOST cautious making a forecast for, and why?
- What additional data would you ask for before giving investment advice?

Is That 'Pattern' Real, or Are You Fooling Yourself?

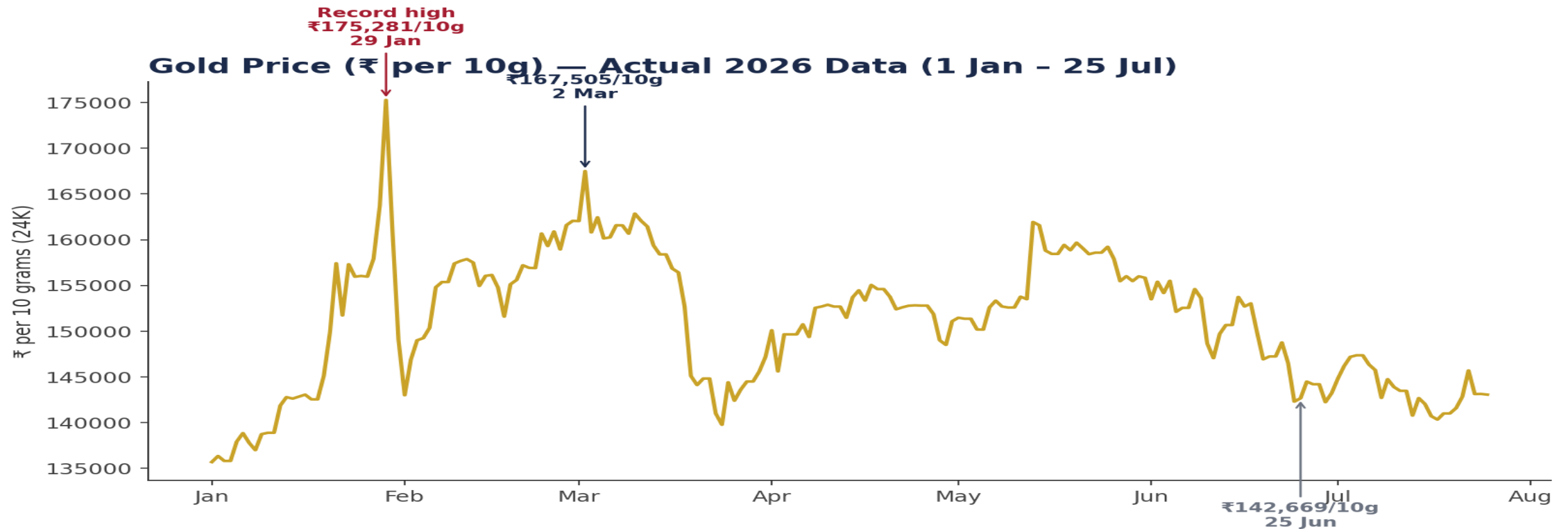
The Trap

- A retail sales manager sees revenue jump every December and concludes 'our new marketing campaign is working.'
- In reality, it might just be a HOLIDAY SEASONALITY effect that happens every year, campaign or not.
- Confusing a structural pattern with a one-off cause is one of the most expensive mistakes in business analytics.

What We Need

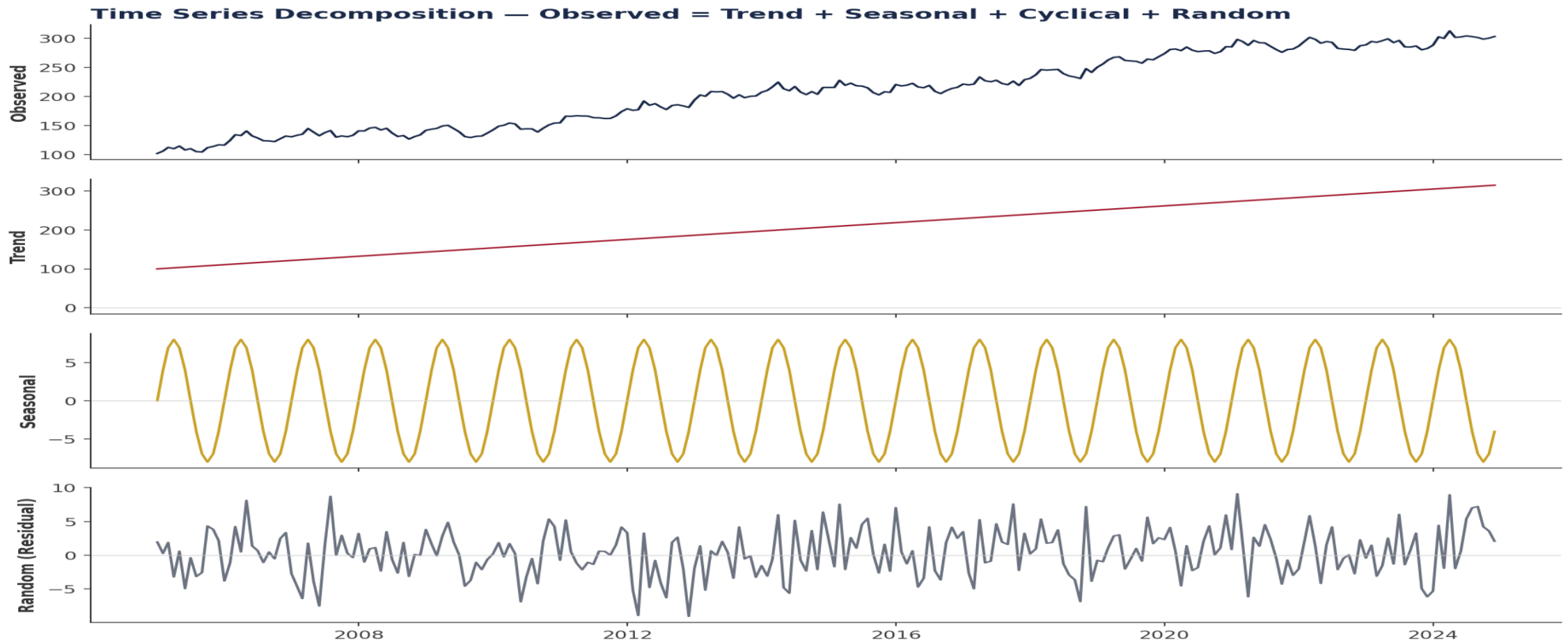
- A systematic way to break any series into its underlying building blocks.
- So we can ask: is this movement TREND, SEASONALITY, CYCLE, or just NOISE?

Trend, Shock, or Noise? Read the Real 2026 Gold Chart



 2 minutes: is this a long-run **TREND**, a one-off **SHOCK** followed by partial reversion, or just **NOISE**? Defend your answer to your partner — you should already have a view from the pre-class reading.

Decomposition — Observed = Trend + Seasonal + Cyclical + Random



The Four Components, Defined

Trend

The long-run direction of the series — persistent growth or decline over years.
Example: gold's multi-decade rise.

Seasonality

A pattern that repeats at FIXED, KNOWN intervals (daily, monthly, quarterly).
Example: retail sales spiking every December.

Cyclical

A pattern that repeats but at IRREGULAR, unpredictable intervals — often tied to the business or credit cycle. Example: multi-year boom-bust cycles.

Random (Residual)

Whatever is left after removing trend, seasonality and cycle — genuine unpredictable noise.

Key distinction: seasonality has a FIXED calendar period (always December); cyclicity does not (the next recession could be in 3 years or 9).

Time-Series Decomposition in Python

Classical Decomposition

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# df has a DatetimeIndex and a 'price' column
result = seasonal_decompose(
    df['price'],
    model='additive', # or 'multiplicative'
    period=12         # 12 = monthly seasonality
)

result.plot()
plt.show()
```

Reading the Output

```
# result.trend      -> long-run direction
# result.seasonal   -> repeating calendar pattern
# result.resid      -> what's left (noise)

# Quick sanity check:
reconstructed = (
    result.trend
    + result.seasonal
    + result.resid
)

# Should closely match the original series
(reconstructed - df['price']).abs().max()
```

Turning Decomposition Into Decisions

Trend

Strategy-Use the trend component for capital budgeting, long-term capacity planning, multi-year forecasts.

Seasonality

Operation- Use the seasonal component for inventory, staffing, and working-capital planning within the year.

Cyclical

Risk Management-Use the cyclical component to time provisioning, stress tests, and counter-cyclical buffers.

Is Market Seasonality Real — or Are We Fooling Ourselves?



Vote

- A. 'Sell in May and go away' is a genuine, exploitable seasonal pattern.
- B. It's a coincidence dressed up as a pattern — noise, not signal.
- C. It may have existed once but has been arbitrated away.



Class Debate (3 min)

- Defend your vote to a neighbor who voted differently.
- What EVIDENCE would change your mind?
- We will return to this question with formal tools by the end of the session.

Why Do Even Expert Analysts Get Forecasts Wrong?

The Uncomfortable Fact

- Professional equity analysts, with Bloomberg terminals, industry contacts, and years of training, routinely miss major market moves.
- Most actively managed equity mutual funds underperform their benchmark index over 10+ year periods.
- This isn't a lack of effort or intelligence — it may be a fundamental property of the data itself.

Today's Central Question

- Is there a mathematical reason WHY prices are hard to predict — not just 'markets are complicated'?
- Spoiler: yes. It's called the Random Walk, and we'll build it from first principles in the next block.

Predict Tomorrow's NIFTY Direction



📝 Write down: will NIFTY close UP or DOWN tomorrow, and your confidence (50-100%). We'll check the actual outcome and tally the class accuracy.

When the Experts Disagree — and Are Both Wrong

Wall Street Year-End Targets

- Every December, major banks publish next-year index targets for NIFTY, S&P 500, etc.
- Historically, the spread between the most bullish and most bearish forecasts is often larger than the ACTUAL annual move.
- Forecasts also cluster around 'more of the same' — extrapolating the recent trend, not genuinely predicting turning points.

The Lesson

- If the smartest, best-resourced forecasters on the planet can't agree — and collectively miss most turning points — that's strong evidence prediction is structurally hard, not just an individual skill gap.

The Efficient Market Hypothesis (Intuition First)

The Core Idea

- If a pattern in prices were reliably predictable and profitable, traders would exploit it immediately.
- That trading activity itself pushes prices to eliminate the pattern — it 'arbitrages itself away.'
- Result: prices should reflect all currently available information almost instantly.

What This Implies

- Only NEW information (news, earnings, policy surprises) should move prices going forward.
- By definition, new information is unpredictable — otherwise it wouldn't be 'new.'
- This is the theoretical bridge to our next topic: the Random Walk.

What Market Efficiency Means for a Manager

For a Portfolio Manager

- Justifies broad diversification and low-cost index investing as a rational default, not a lack of ambition.
- Active management can still add value through superior RISK management, not necessarily superior prediction.

For a Corporate Strategist

- You cannot reliably time equity issuance or buybacks based on 'the stock is cheap right now.'
- Focus capital-structure decisions on fundamentals and financing needs, not on market-timing bets.

If Prices Are Unpredictable, What Model Fits Them?



The Challenge

- We've argued prices should reflect all known information, so CHANGES in price should be unpredictable.
- But we still need SOME mathematical model — 'I don't know' isn't good enough for a risk model.
- We need a model whose defining feature IS unpredictability.



Enter the Random Walk

- The single most important model in financial time series.
- It says: tomorrow's price = today's price + a completely unpredictable random shock.
- Deceptively simple — and it explains an enormous amount of real market behavior.

Is a Coin Flip Like a Stock Price?

The Setup

- Start a counter at 0. Flip a coin: heads = +1, tails = -1.
- Plot the running total after each flip.
- Do this for 3 minutes with a partner (mentally or on paper) — 20 flips.

Discuss

- Does your running total ever look like it has a 'trend,' even though each flip is pure 50/50 chance?
- Could you predict flip #21 from flips #1-20?
- How is this similar to — and different from — a stock price?

Random Walk WITHOUT Drift

$Y_t = Y_{t-1} + \varepsilon_t$, where ε_t is unpredictable 'white noise' (mean 0)

In Plain English

- Today's value equals yesterday's value, PLUS a completely random shock.
- The best forecast for tomorrow is simply today's value — nothing else helps.
- There is no pull back to any long-run average — the series can wander anywhere.

Key Properties

- $E[Y_t] = Y_0$ (no expected direction).
- Variance GROWS with time: $\text{Var}(Y_t) = t \cdot \sigma^2$ — the longer you wait, the wider the range of possible outcomes.
- This growing uncertainty is exactly why long-horizon forecasts are so much less precise than short-horizon ones.

Simulating a Random Walk Without Drift

Simulate 5 Paths

```
import numpy as np
import matplotlib.pyplot as plt

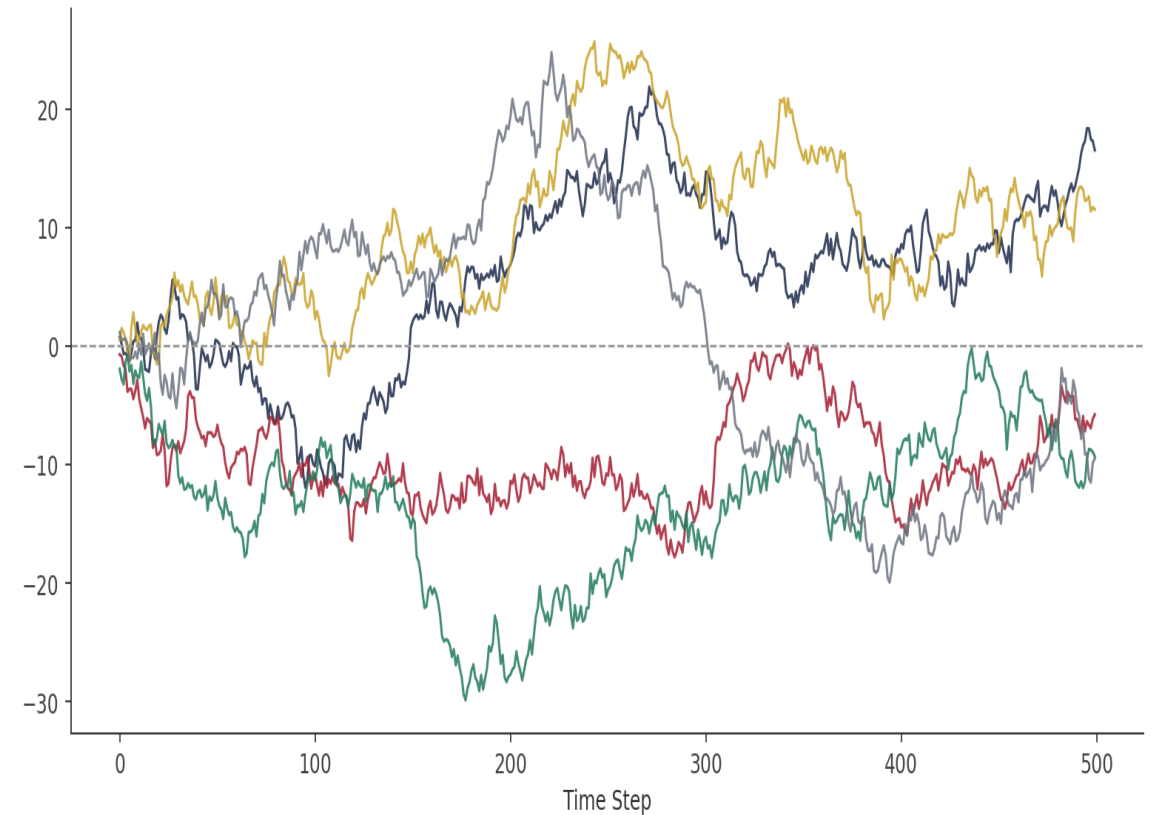
np.random.seed(42)
n_steps = 500
n_paths = 5

# each step: random shock, mean 0, sd 1
shocks = np.random.normal(0, 1, (n_paths, n_steps))

# cumulative sum = the random walk itself
paths = np.cumsum(shocks, axis=1)

for p in paths:
    plt.plot(p)
plt.axhline(0, color='black', ls='--')
plt.title('Random Walk Without Drift')
plt.show()
```

Random Walk WITHOUT Drift — Five Simulated Paths ($Y_t = Y_{t-1} + \varepsilon_t$)



Random Walk WITH Drift

$Y_t = \mu + Y_{t-1} + \varepsilon_t$, where μ is a constant drift term

+ What Changed

- We add a constant μ ('mu') to every step — a systematic push in one direction, on top of the randomness.
- $\mu > 0$: a persistent upward bias (like long-run equity returns, driven by earnings growth and inflation).
- $\mu < 0$: a persistent downward bias (e.g., a currency under sustained depreciation pressure).

Why This Matters for Equities

- This is the best simple explanation for why stock indices trend upward over decades DESPITE being unpredictable day-to-day.
- The drift is small per day but compounds into a large effect over years — exactly like NIFTY's long-run rise.

Simulating a Random Walk With Drift

Add a Drift Term

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
n_steps = 500
mu = 0.35 # drift per step

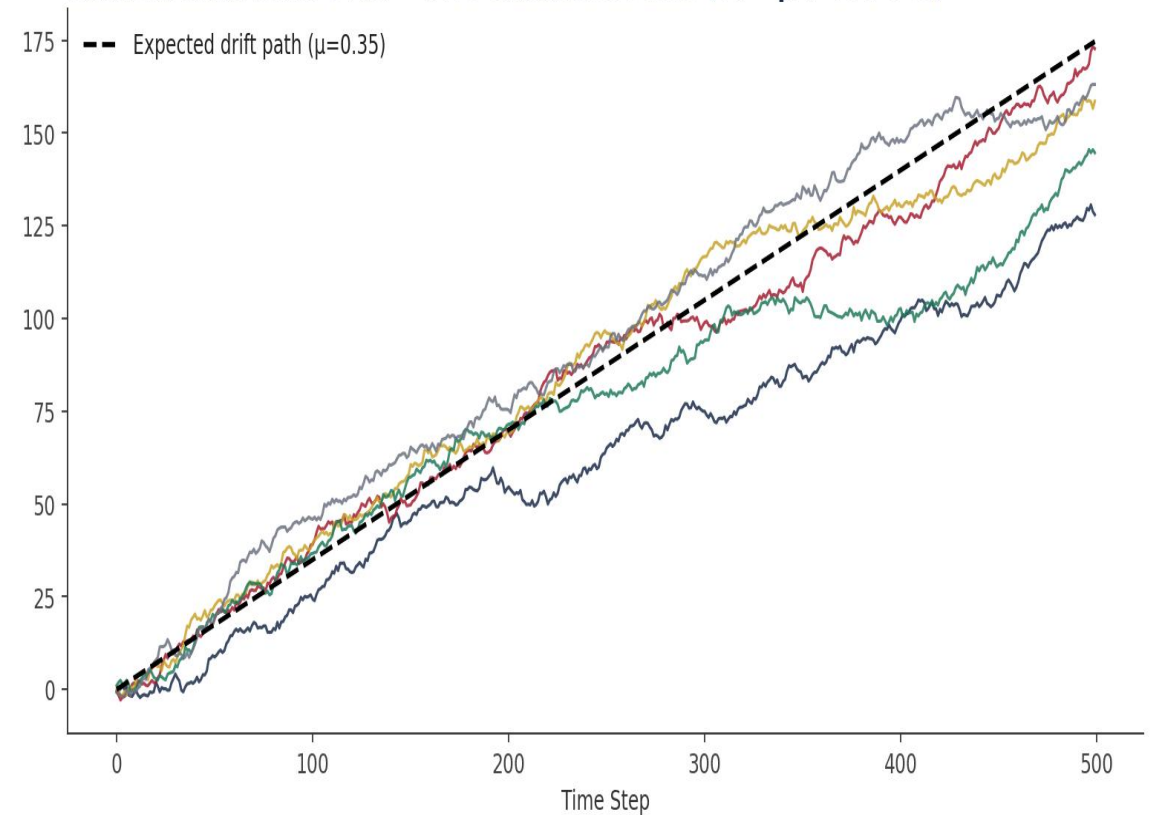
shocks = np.random.normal(0, 1, (5, n_steps))

# add drift BEFORE cumulative sum
paths = np.cumsum(mu + shocks, axis=1)

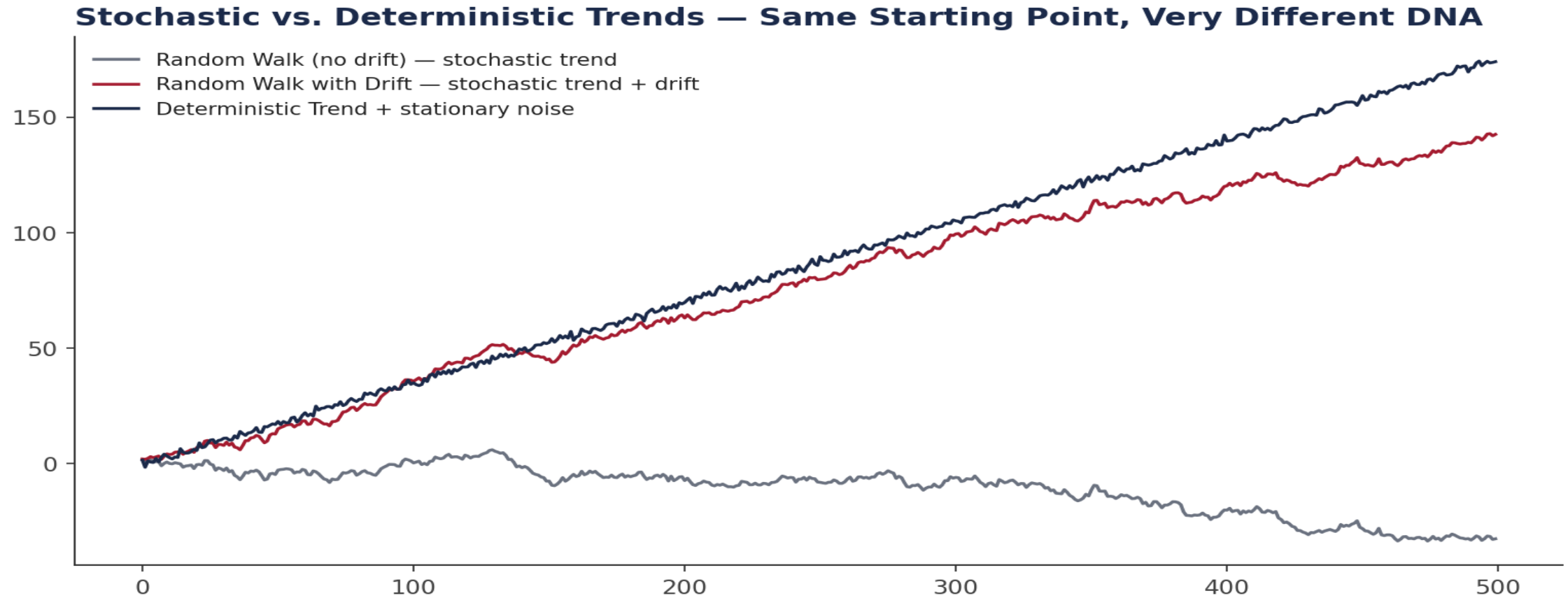
expected_path = mu * np.arange(n_steps)

for p in paths:
    plt.plot(p, alpha=0.85)
plt.plot(expected_path, 'k--',
          label=f'Expected drift (mu={mu})')
plt.legend()
plt.show()
```

Random Walk WITH Drift — Five Simulated Paths ($Y_t = \mu + Y_{t-1} + \varepsilon_t$)



Stochastic vs. Deterministic Trends — Same Start, Different DNA



Unit Root Process — When Shocks Are Permanent



The Core Idea

- A series 'has a unit root' if a shock today shifts the entire future path of the series permanently — it never fades.
- The random walk is the textbook example of a unit-root process (coefficient on Y_{t-1} is exactly 1).
- Contrast: a mean-reverting series eventually forgets shocks and returns to its average.



Why This Matters

- A market crash in a unit-root series (like a stock price) permanently resets the 'new normal' level.
- A shock in a mean-reverting series (like an interest-rate spread) is temporary — expect a snap-back.
- This is exactly the diagnostic question analysts formally test with a 'unit root test' (e.g., Augmented Dickey-Fuller) before building any forecasting model.

Stochastic Trend vs. Deterministic Trend



Deterministic Trend

- The series follows a fixed mathematical line PLUS temporary, fading noise.
- Example mental model: a factory's output capacity growing on a fixed schedule, with random weekly hiccups that don't compound.
- Fix: 'de-trend' by subtracting the fitted line — what's left is stationary.



Stochastic Trend

- The trend itself is built from accumulated RANDOM shocks — a random walk (with or without drift), like most asset prices.
- There is no fixed line to 'subtract' — each shock permanently changes the future path.
- Fix: 'difference' the series (look at period-to-period CHANGES) instead of de-trending.

What Random Walk Theory Means for Forecasting and Risk

Forecasting

- For a unit-root price series, the best point forecast for any future date is simply today's price (plus drift, if any).
- Elaborate 'chart pattern' forecasts add little beyond this simple baseline — a healthy benchmark to test any model against.

Risk Management

- Because variance grows with time ($\text{Var} \propto t$), risk (e.g., Value-at-Risk) should scale with the square root of the time horizon — the famous 'square-root-of-time rule.'
- This is precisely why long-horizon financial forecasts carry enormous uncertainty bands.

Can Technical Analysis Beat a Random Walk?



Team A: Technical Analysis Works

- Chart patterns reflect real crowd psychology and order-flow information.
- Momentum and mean-reversion strategies have published, peer-reviewed evidence of (modest) historical returns.



Team B: It's a Random Walk

- Any real pattern gets arbitrated away once well known (Efficient Market Hypothesis, Module 4).
- Most documented 'anomalies' shrink or disappear out-of-sample and after costs.

 5-minute structured debate: split the room, 90 seconds per side, then 2 minutes of rebuttal. No resolution required — just sharpen your reasoning.

Why Can't We Just Run a Regression on Raw Stock Prices?

The Naive Approach

- A junior analyst regresses HDFC Bank's stock price on Reliance's stock price and gets a beautiful $R^2 = 0.92$.
- They conclude the two are 'strongly related' and build a pairs-trading strategy on it.
- The strategy loses money immediately. What went wrong?

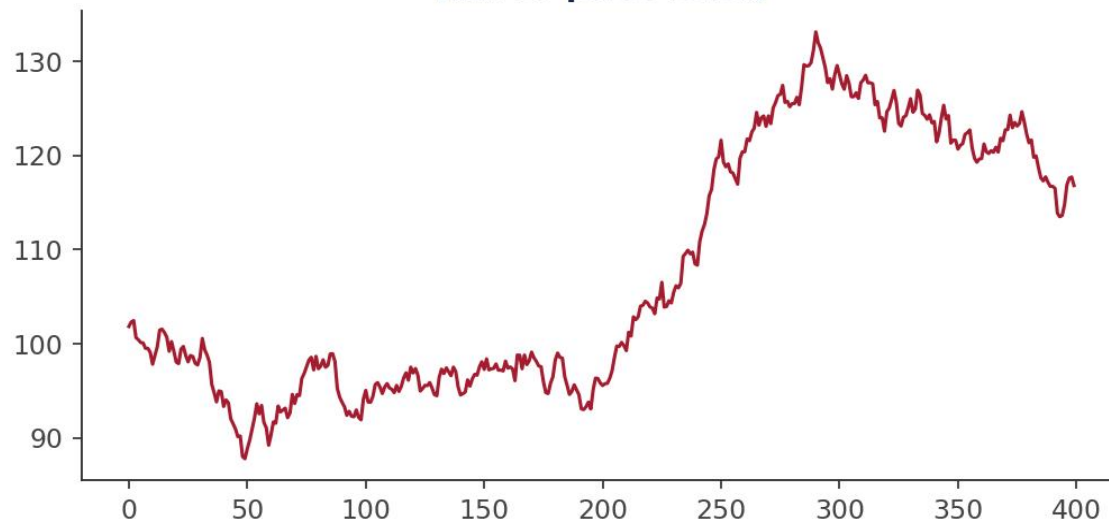
The Hidden Problem

- Two UNRELATED trending series will almost always show a high R^2 — this is called SPURIOUS REGRESSION.
- It happens because both series are non-stationary — their means and variances change over time, breaking the assumptions behind ordinary regression.

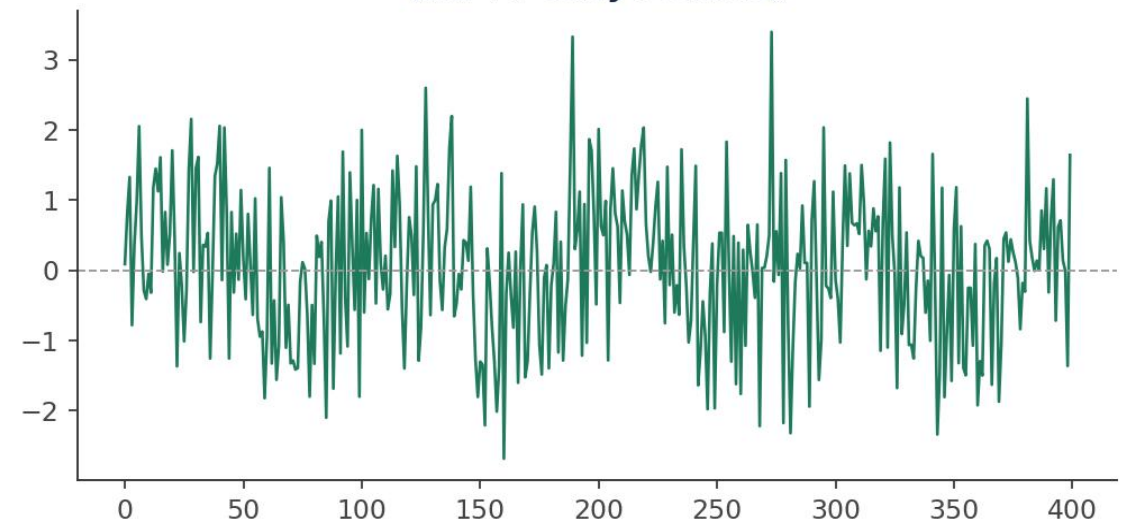
Which Chart Is Safe to Model Directly?

Stationary vs. Non-Stationary — Spot the Difference

**Non-Stationary
(NIFTY price level)**

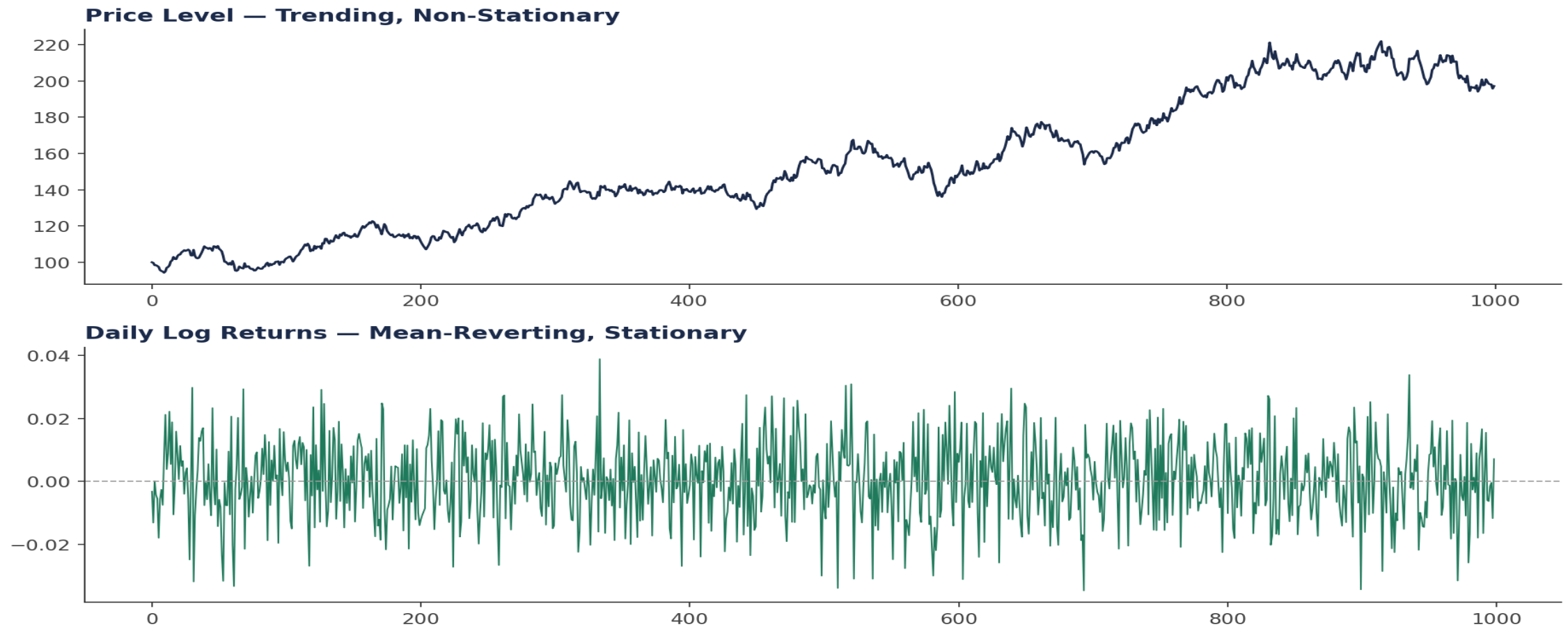


**Stationary
(NIFTY daily returns)**



 2 minutes: one of these is stationary, one is not. Which is which, and what specifically tells you?

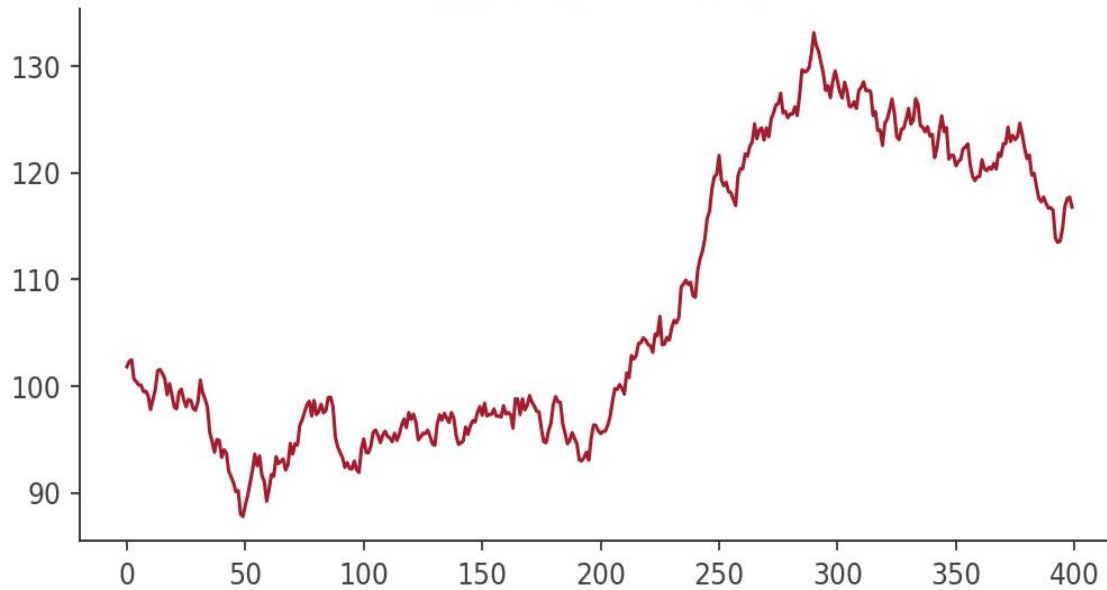
NIFTY Price Level vs. NIFTY Daily Returns



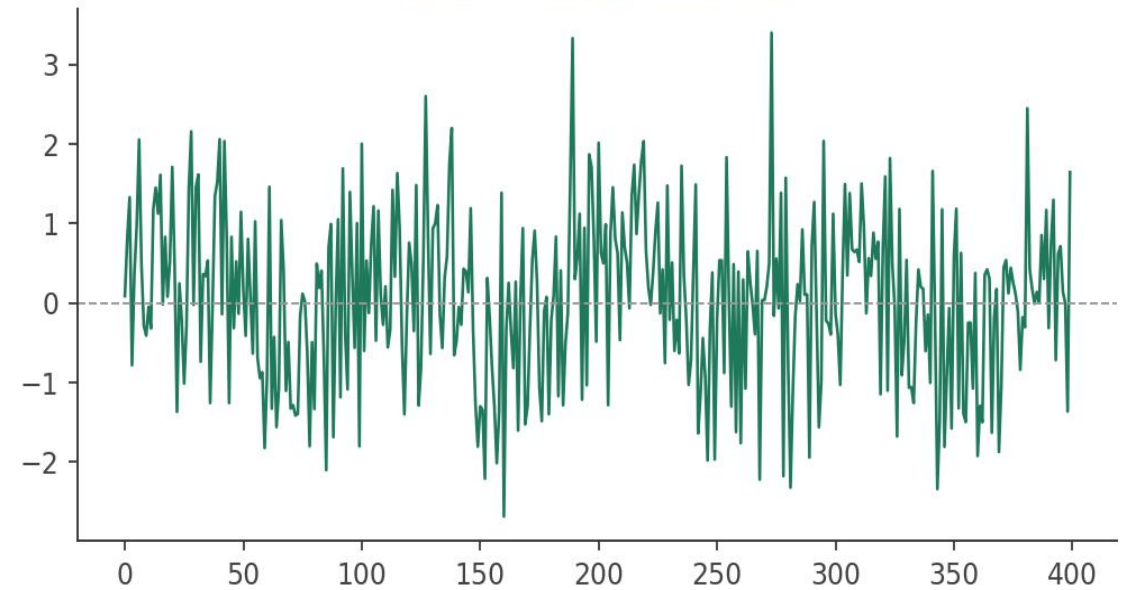
Stationary vs. Non-Stationary, Side by Side

Stationary vs. Non-Stationary — Spot the Difference

**Non-Stationary
(NIFTY price level)**



**Stationary
(NIFTY daily returns)**



Stationarity — A Formal Definition, Built From Intuition



Three Conditions

- 1. Constant mean: the series doesn't systematically drift up or down over time.
- 2. Constant variance: the 'spread' of the series doesn't grow or shrink over time.
- 3. Constant autocorrelation: the relationship between today and yesterday doesn't change depending on WHEN you look.



Why It Matters

- Most statistical models (regression, ARIMA, correlation) implicitly assume these three properties hold.
- Violate them, and your standard errors, p-values, and R^2 can all be dangerously misleading — exactly the spurious regression trap from Slide 36.

Visualizing Stationarity With Rolling Statistics

Rolling Mean & Std Dev

```
import pandas as pd
import matplotlib.pyplot as plt

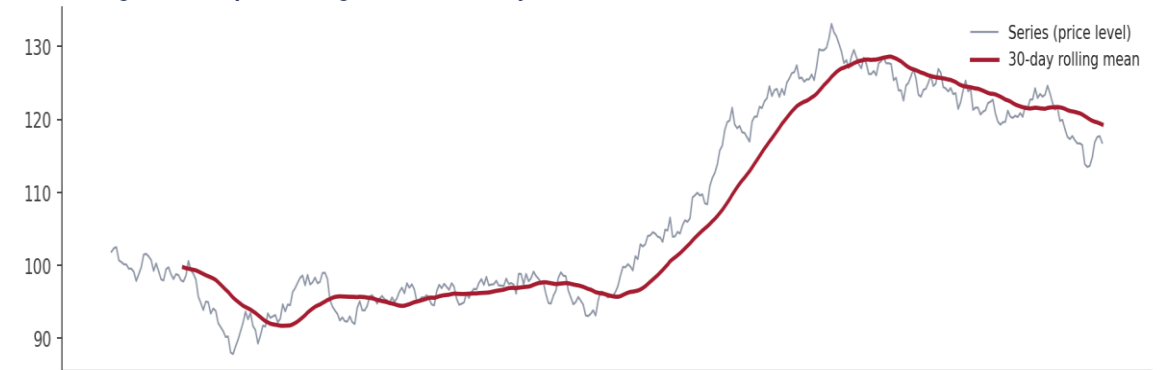
roll_mean = series.rolling(window=30).mean()
roll_std = series.rolling(window=30).std()

fig, ax = plt.subplots(2, 1, sharex=True)
ax[0].plot(series, alpha=0.4, label='Series')
ax[0].plot(roll_mean, label='30-day mean')
ax[0].legend()

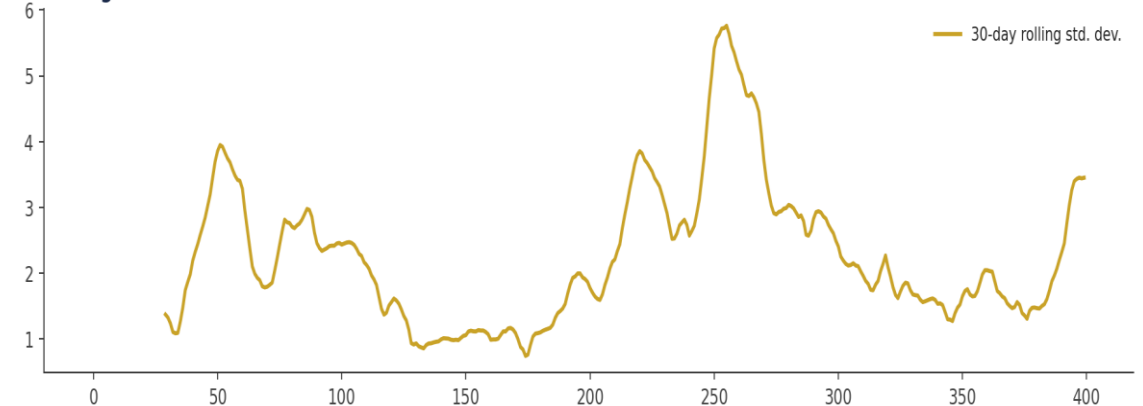
ax[1].plot(roll_std, label='30-day std dev')
ax[1].legend()
plt.show()

# Formal test (not required for MBA level):
# from statsmodels.tsa.stattools import adfuller
# adfuller(series) # p < 0.05 -> reject unit root
```

Rolling Mean Keeps Climbing → Non-Stationary



Rolling Std. Dev. Also Drifts



Why Stationarity Is a Pre-Flight Checklist, Not Trivia

✓ Before You Model Anything

- Always plot the series and check: does the mean wander? Does the spread change over time?
- If EITHER answer is yes, transform the data (e.g., use returns instead of prices) before regressing, correlating, or forecasting.

✖ The Cost of Skipping This Step

- Spurious 'relationships' that don't hold up out-of-sample.
- Overconfident risk models that understate real uncertainty.
- Client-facing analysis that looks rigorous but is statistically invalid.

Why Are Stock Prices Usually Non-Stationary?

The Mechanism

- A price is a CUMULATIVE SUM of all past returns — remember, $P_t = P_0 \times (1 + r_1)(1 + r_2)\dots(1 + r_t)$.
- Because prices accumulate every past shock (recall the unit root property from Module 5), they have no fixed level to return to.
- Each day's closing price becomes the new baseline — there's no 'gravity' pulling it back.

The Practical Symptom

- The mean of a price series taken over the first half of history is almost never equal to the mean over the second half.
- This is precisely condition 1 of stationarity (constant mean) failing.

Why Returns Are Preferred — The Power of Differencing

÷ The Fix: Differencing

- Instead of modeling the LEVEL (price), model the CHANGE (return): $r_t = (P_t - P_{t-1}) / P_{t-1}$, or log return = $\ln(P_t/P_{t-1})$.
- This single transformation removes the accumulated unit root and typically produces a stationary series.
- It's the standard first step in almost every financial time-series model you'll encounter in industry.

⚖ Bonus: Comparability

- Returns are also SCALE-FREE — you can directly compare a ₹1,500 stock's return to a ₹15 stock's return, which raw prices never allow.
- This is why every performance chart you've seen in finance (fund NAV growth, index returns) is really built from returns, not raw price differences.

Comparing Prices and Returns

Compute & Plot Both

```
import numpy as np
import matplotlib.pyplot as plt

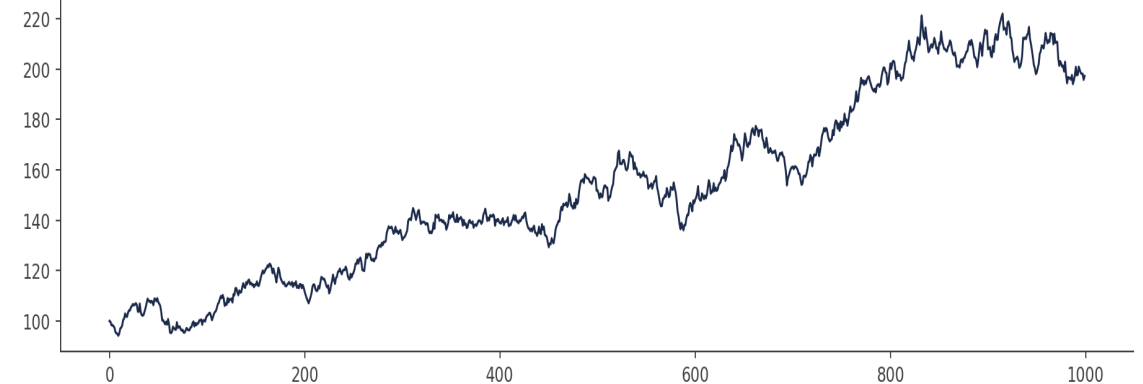
price = df['close']

# simple returns
returns = price.pct_change().dropna()

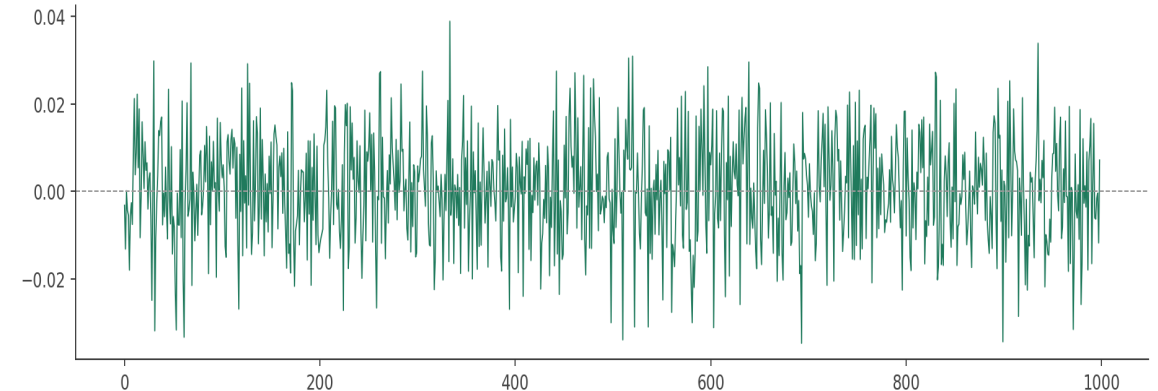
# log returns (preferred – additive over time)
log_returns = np.log(price / price.shift(1)).dropna()

fig, ax = plt.subplots(2, 1, sharex=False)
ax[0].plot(price)
ax[0].set_title('Price Level')
ax[1].plot(log_returns)
ax[1].axhline(0, ls='--', color='grey')
ax[1].set_title('Log Returns')
plt.show()
```

Price Level – Trending, Non-Stationary



Daily Log Returns – Mean-Reverting, Stationary



You Are a Portfolio Manager Building a Risk Model



The Brief

- Your risk team must estimate the probability that the portfolio loses more than 5% in a single day (Value-at-Risk).
- An intern proposes fitting a statistical distribution directly to the last 3 years of DAILY PRICE LEVELS.



Discuss in Pairs (3 min)

- Is the intern's approach sound? Why or why not?
- What should the risk model be built on instead?
- What data structure question (Module 2) is also relevant here if the portfolio holds 30 stocks?

The Golden Rule: Model Returns, Report Prices

Golden rule: BUILD your statistical models on returns (stationary); COMMUNICATE results back to stakeholders in prices or rupee terms (intuitive).



Build On Returns

- Risk models (VaR, volatility forecasting), correlation and regression analysis, portfolio optimization.



Report In Prices / ₹

- Client statements, board presentations, P&L — nobody wants to see 'expected log return -0.014' on a client report.

Prices or Returns — Which Would You Model, and Why?

Vote

- A. Prices — they're more intuitive and directly meaningful.
- B. Returns — they're stationary and statistically valid to model.
- C. It depends on the specific business question.

Justify Your Vote (2 min)

- In one sentence, justify your answer to a partner using a term from today's session (stationarity, unit root, drift).

Business Implications — Where This Toolkit Gets Used

Forecasting

Sales, demand, and cash-flow forecasting all rely on correctly separating trend/seasonality from noise.

Risk Management

VaR, stress testing, and capital buffers depend on understanding how uncertainty grows over time.

Strategy & Investment

Deciding whether a market move is a real regime change or expected noise shapes M&A timing and capital allocation.

The RBI Governor Asks: 'Is Inflation a Trend or a Cycle?'

The Brief

- Inflation has been above the RBI's 6% upper tolerance band for 4 straight months.
- The Monetary Policy Committee must decide: raise rates aggressively (treating this as a persistent trend), or hold steady (treating it as a temporary cyclical spike, e.g., from a one-off food-price shock)?

Discuss in Pairs (3 min)

- What data would you want to see to distinguish trend from cycle here (Module 3)?
- What's the cost of being WRONG in each direction — raising rates on a temporary spike, vs. holding on a real trend?

Write a One-Slide Investment Memo

The Brief

- You are an equity analyst. A client asks whether to increase their allocation to gold given its 5-year trend (Slide 14).
- Using ONLY today's vocabulary (trend, stationarity, drift, random walk), draft 3 bullet points for your recommendation memo.

Individually (3 min), Then Share

- Bullet 1: what the historical TREND tells you (and doesn't).
- Bullet 2: how confident can you be extrapolating that trend forward, given random-walk behavior?
- Bullet 3: your recommendation, with an explicit caveat about uncertainty.

Decision Tree: Choosing the Right Technique

1. Plot the raw series

Always start here. Does the mean or spread visibly change over time?

2. Decompose (Module 3)

Separate trend, seasonality, cycle, and noise. Is the movement structural or random?

3. Test stationarity (Module 6)

Check rolling mean/variance, or run a formal test (e.g., ADF). Stationary? Proceed. Not stationary? Difference it.

4. Model returns, not prices

Build your statistical model (regression, correlation, forecasting) on the stationary, differenced series.

5. Translate back to business terms

Report results in ₹, %, or index points — never leave a client with a raw statistical output.

Rolling & Moving Averages — Smoothing Out the Noise

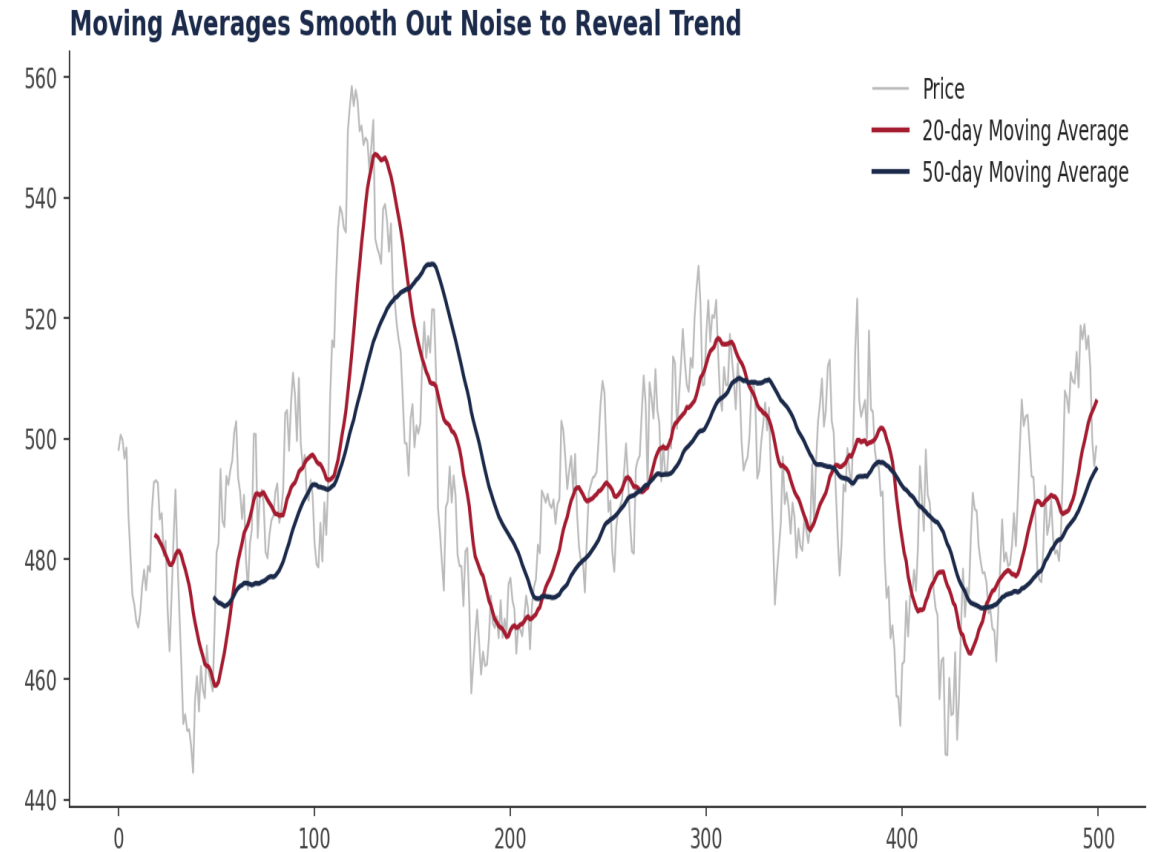
Compute Moving Averages

```
import pandas as pd
import matplotlib.pyplot as plt

price = df['close']

ma20 = price.rolling(window=20).mean()
ma50 = price.rolling(window=50).mean()

plt.plot(price, alpha=0.35, label='Price')
plt.plot(ma20, label='20-day MA')
plt.plot(ma50, label='50-day MA')
plt.legend()
plt.title('Moving Averages Smooth Short-Term Noise')
plt.show()
```



Build and Defend a Simple Forecasting Model

In Groups of 4 (8 minutes)

- Choose ONE series from today's dashboard (Slide 10): Reliance, HDFC Bank, USD/INR, Gold, Bitcoin, or Crude Oil.
- Propose a simple forecasting approach (e.g., 'random walk with drift', 'moving average', 'seasonal + trend').
- State ONE assumption your approach relies on, and ONE way it could fail.

Then: Challenge Another Team

- Swap your approach with a neighboring group.
- Find ONE weakness or hidden assumption in their proposed model.
- Report back: what would you tell them to check before trusting their forecast?

You Are the Analytics Team for a New Client



The Engagement

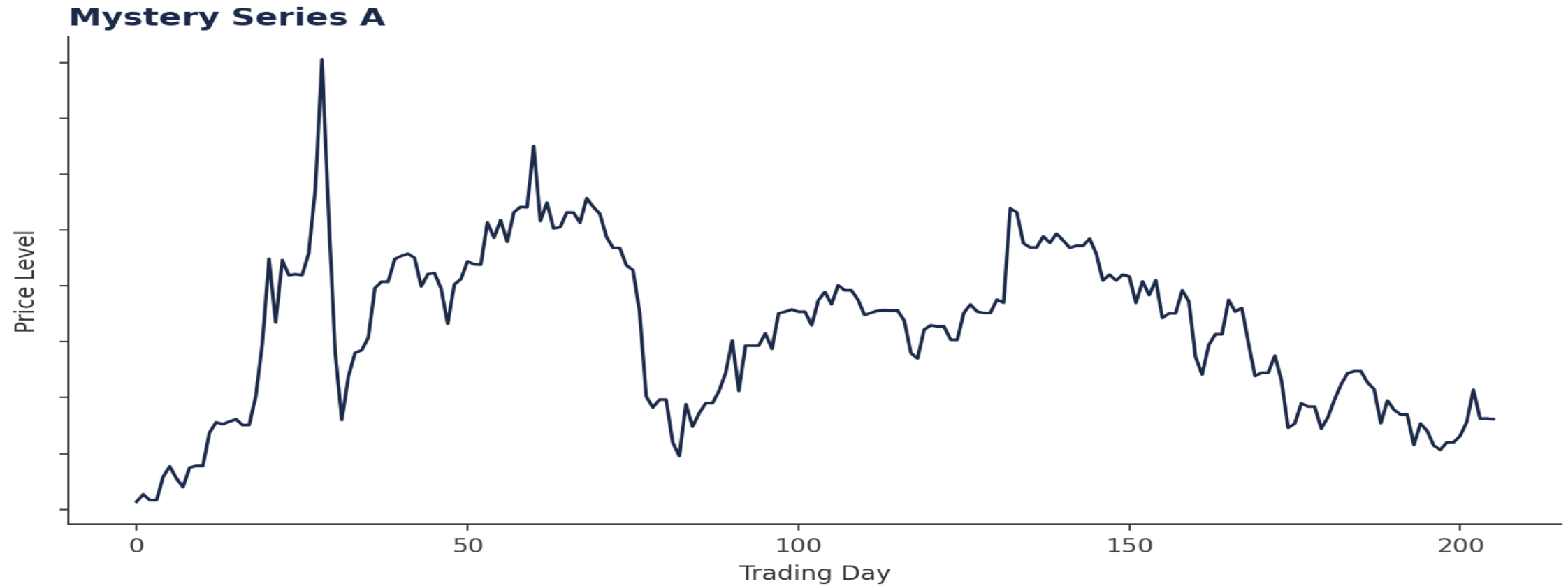
- A client has handed you three unlabeled financial time series (A, B, C) and asked for a written diagnostic memo before any forecasting begins.
- You do NOT get to see what each series represents — only the data itself, exactly like a real first-pass diagnostic.



Format

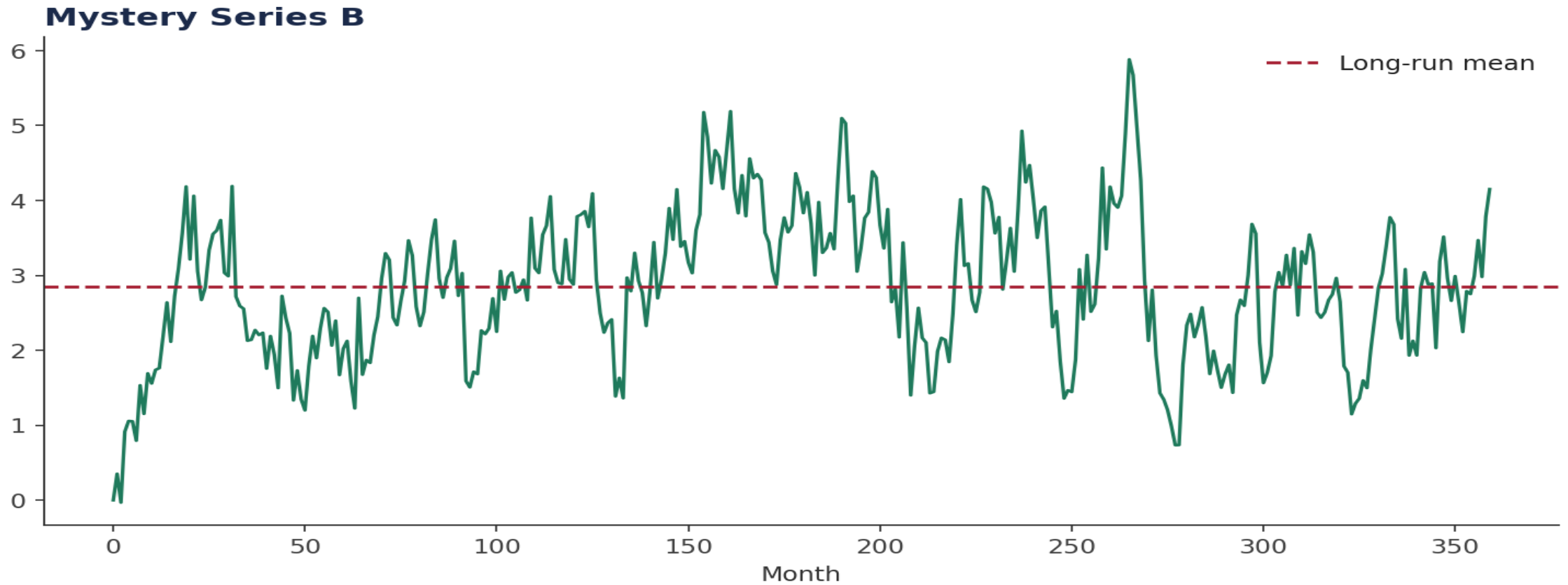
- Work in your existing groups of 4.
- You will see all three charts on the next slides, then receive your written task.
- No answers will be revealed today — you will defend your memo to the class.

Mystery Series A



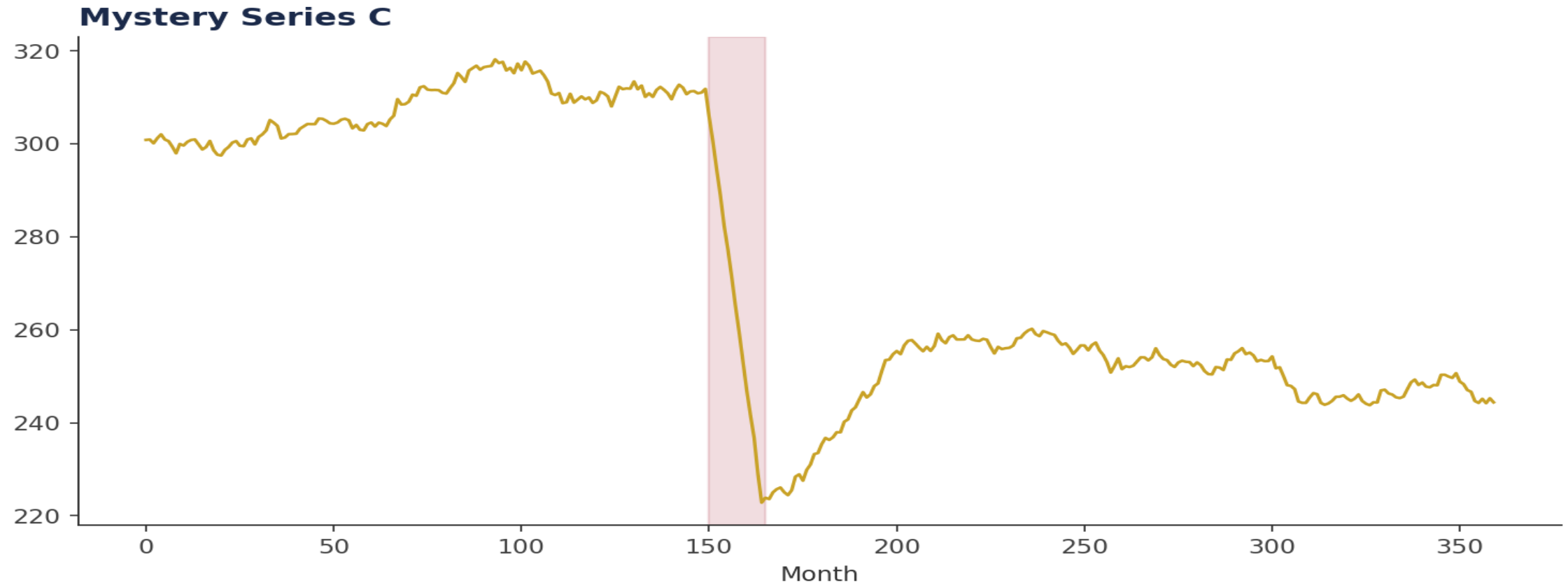
No labels, no hints. Just observe for now — your formal task instructions come after Series C.

Mystery Series B



No labels, no hints. Just observe for now — your formal task instructions come after Series C.

Mystery Series C



No labels, no hints. Now that you've seen all three, let's read your formal task.

Diagnostic Memo — Due at End of Session



For EACH of the Three Series, Answer:

- 1. Identify the components present (trend? seasonality? cyclical? a structural break?).
- 2. Does this series appear STATIONARY or NON-STATIONARY? What specific evidence supports your answer?
- 3. Would you model the PRICE (level) or the RETURN (change) — and why?
- 4. Justify your reasoning in business language a client could understand — not just statistical jargon.



Logistics

- 15 minutes in your group of 4.
- One page maximum per series — bullet points, not paragraphs.
- Each group presents ONE series (assigned) to the class in 90 seconds.
- No answers are provided today — we will discuss reasoning quality, not a single 'correct' label.

15 Minutes on the Clock — Assign Roles

Suggested Roles

- Chart Reader: describes what's visually observed, no jargon yet.
- Statistician: translates observations into stationarity/trend language.
- Business Translator: converts the analysis into a client-ready recommendation.
- Presenter: delivers the 90-second readout.

Remember

- Use the 5-step decision tree from Slide 52 as your checklist.
- There is no single 'correct' answer being hidden — we are grading REASONING QUALITY.
- Disagreement within your group is a feature, not a bug — capture both views if you can't agree.

Six Ideas to Take With You

1. Order Matters

Time series data cannot be shuffled — today depends on yesterday. This is what makes the whole field different from cross-sectional analysis.

2. Decompose Before You Conclude

Separate trend, seasonality, cycle, and noise before crediting any strategy or event with causing a movement.

3. Markets Are (Mostly) a Random Walk

Prices largely reflect a random walk, often with drift — the best simple forecast for tomorrow is close to today, plus a small trend.

4. Stationarity Is a Pre-Flight Check

Always check whether mean and variance are stable before modeling — or risk spurious, misleading results.

5. Model Returns, Report Prices

Build your statistics on stationary returns; translate results back into intuitive price/rupee terms for stakeholders.

6. Prediction Has Real Limits

Uncertainty grows with time — humility about long-horizon forecasts is a feature of good analysis, not a weakness.

Cheat-Sheet: Concept → Python → Business Use

Concept	Python Tool	Business Use
Decomposition	<code>seasonal_decompose()</code>	Separate strategy impact from seasonal noise
Random walk check	<code>np.cumsum(shocks)</code>	Sanity-check whether a 'pattern' beats a naive forecast
Stationarity check	<code>rolling().mean()/std()</code>	Pre-flight check before any regression or correlation
Returns	<code>.pct_change() / log()</code>	The correct series to model, forecast, or risk-manage
Moving averages	<code>rolling().mean()</code>	Smooth noise to spot the underlying trend

Keep this slide — it maps every statistical idea from today to one line of Python and one business use case.

Debunking What We Often Get Wrong

✗ Myth

- 'A random walk means the market is completely unpredictable in every sense.'
- 'High R^2 means two series are truly related.'
- 'If a series looks noisy and jumpy, it must be non-stationary.'

✓ Reality

- A random walk WITH drift still has a predictable long-run direction — only the short-term wiggle is unpredictable.
- Two independently trending (non-stationary) series can show high R^2 with zero real relationship — spurious regression.
- A noisy, jumpy series can still be stationary (constant mean/variance) — stationarity is about STABILITY of statistical properties, not smoothness.

Would the Answer Change During COVID?

The Question

- Everything we discussed today — drift, stationarity, decomposition — was built mostly on 'normal times' behavior.
- During COVID, correlations, volatility, and even the drift term itself shifted dramatically and suddenly.

Discuss (3 min)

- Does a single structural break like COVID mean our models are useless — or just that they need re-estimation?
- How would you, as an analyst, detect that 'the rules changed' in near real time rather than months later?
- Is correlation ever 'enough' to act on, or do you always need a causal story too?

SESSION WRAP-UP

“Can Anyone Beat the Stock Market?” — has your answer from Slide 3 changed?

TODAY YOU LEARNED

To separate trend, seasonality, and noise — and to test before you trust a pattern.

NEXT SESSION

ARIMA models, forecasting evaluation, and volatility clustering (GARCH intuition).

BEFORE NEXT CLASS

Bring one real time series from your own industry — we'll diagnose it together.